



Quantification and potential functional relevance of binding cooperativity of adjacent transcription factors on DNA

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In eukaryotes, the expression of specific genes is regulated by a combination of transcription factors (TFs) bound on regulatory regions of the genomic DNA (promoters and enhancers). Recent advances in genomic sequencing technology have enabled the measurements of TFs' footprints and binding affinities on DNA at the single-molecule level, facilitating the probing of binding cooperativity among adjacent TFs. This necessitates quantitative descriptions of TFs' binding cooperativity and understanding of its potential functional relevance. In this study, we show that the binding cooperativities between two adjacent TFs can be quantified by the γ coefficient, which can be experimentally determined. Under thermodynamic equilibrium, the binding affinities of two TFs can either increase together (positive cooperativity) or decrease together (negative cooperativity), but not in opposing directions (one increases while the other decreases). Within the framework of thermodynamics, we investigate the functional relevance of cooperativity. The functional relevance of positive cooperativity, which has been extensively discussed in the literature, is the sigmoidal binding curve around a TF concentration threshold (analogous to oxygen binding to hemoglobin), whereas the functional relevance of negative cooperativity is twofold. First, mutual exclusion of the two TFs enables bidirectional gene switching, akin to the CI-Cro system in phage λ . Second, while TFs often exhibit intranuclear concentration fluctuations, negative binding cooperativity assures fast TF dissociation from DNA and hence rapid response for gene expression regulation. Furthermore, the nonequilibrium steady states of living cells can lead to either positive or negative cooperativity, which can also be quantified by the γ coefficient.

transcription factor | binding affinity | negative cooperativity | functional relevance | nonequilibrium steady state

Binding cooperativity among biomolecules is fundamental to many biological processes, and the classic example is the binding of multiple oxygen molecules to hemoglobin (1). Binding cooperativity can be either positive or negative, the former is exemplified by hemoglobin's oxygen binding (1), and the latter has received increased attention (2–4), though its functional relevance remains elusive. Multiple transcription factors (TFs) bind to DNA regulatory regions, such as promoters and enhancers, to control gene expression (5–7). Mechanistically, positive TF binding cooperativity arises from protein–protein interactions among TFs (8–10), which recruits multiple proteins to assemble functional complexes. Negative TF binding cooperativity may result from competitive binding of TFs to closely spaced binding sites on DNA (11). Furthermore, allostery through DNA can lead to either positive or negative binding cooperativity, which is dependent on the distance between the TFs, and varies with a periodicity of ~ 10 base pairs (12, 13).

The scenarios described above primarily consider the equilibrium binding of TFs to DNA. However, in reality, cells undergo a nonequilibrium steady state, where ATP, ADP, and other cellular metabolites drive dynamic processes (14). Under these nonequilibrium steady-state conditions, TFs, along with other regulatory proteins such as nucleosomes and remodeling complexes (15), engage in dynamic cooperative interactions (16). These interactions may be dynamically modulated by cellular energy fluxes and the fluctuating environment within the nucleus.

Traditionally, TF binding cooperativity can be detected by biochemical bulk measurements such as electrophoretic mobility shift assay (EMSA) (17, 18). Recently, single-molecule imaging (12, 13) and high-throughput sequencing techniques such as CAP-SELEX (19), single molecule footprinting (SMF) (20), Fiber-seq (21), and footprinting with deaminase (FOODIE) (7) offer the promise of detecting TF binding cooperativity with improved accuracy. A key challenge thus arises: How do we quantify TF binding cooperativity using the single-molecule data from these techniques?

Significance

In eukaryotes, multiple transcription factors (TFs) bind to regulatory DNA regions to control gene expression, where adjacent TF binding exhibits positive or negative cooperativity (enhancing or weakening each other's binding). We first present a metric for quantifying such cooperativity based on experimental data under thermodynamic equilibrium. Functionally, positive cooperativity ensures sensitive responses around a TF concentration threshold, whereas negative cooperativity may serve two functions: first, it enables mutual exclusion of TF binding, facilitating bidirectional gene switching; second, as TF concentrations oscillate, negative cooperativity ensures rapid TF dissociation, allowing quick transcriptional switching. Furthermore, the nonequilibrium steady states of living cells can induce either positive or negative cooperativity, which can also be quantified by the metric.

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Moreover, what is the functional relevance of positive and negative binding cooperativity. How do we understand binding cooperativity under nonequilibrium steady-state conditions? This paper addresses these critical questions.

Results

Quantification of TF Binding Cooperativity with the γ Coefficient. Sequencing techniques that detect TF binding at the single-molecule level are typically applied either to a group of cells (in vivo experiments) or to a mixture of purified DNA molecules together with one or a few TFs (in vitro experiments). In both cases, the initial step is to identify TF binding sites at the ensemble level, named as TF footprinting. Subsequently, the TF binding cooperativity between adjacent sites can be statistically quantified as follows.

For any two adjacent TF binding sites, the numbers of DNA molecules cobound (a), either bound (b or c) and both unbound (d) are counted (Fig. 1A). Then, the four conditional probabilities, representing the binding fractions of Site 1 given Site 2 bound and unbound, and the binding fractions of Site 2 given Site 1 bound and unbound, are calculated respectively as

$$\begin{aligned} P_{1b} &= P(\text{Site 1 bound} | \text{Site 2 bound}) = \frac{a}{a+c}, \\ P_{1u} &= P(\text{Site 1 bound} | \text{Site 2 unbound}) = \frac{b}{b+d}, \\ P_{2b} &= P(\text{Site 2 bound} | \text{Site 1 bound}) = \frac{a}{a+b}, \\ P_{2u} &= P(\text{Site 2 bound} | \text{Site 1 unbound}) = \frac{c}{c+d}. \end{aligned} \quad [1]$$

When Site 1 and Site 2 are independent, that means there is no cooperativity, then:

$$\begin{aligned} P_{1b} = P_{1u} &\Leftrightarrow \frac{a}{a+c} = \frac{b}{b+d} \Leftrightarrow \frac{ad}{bc} = 1, \\ P_{2b} = P_{2u} &\Leftrightarrow \frac{a}{a+b} = \frac{c}{c+d} \Leftrightarrow \frac{ad}{bc} = 1. \end{aligned} \quad [2]$$

Here, we define:

$$\gamma \stackrel{\text{def}}{=} \frac{ad}{bc}. \quad [3]$$

Thus, $\gamma = 1$ is necessary and sufficient for no cooperativity.

When Site 1 and Site 2 are dependent, there are two opposite situations. If Site 1 and Site 2 are positively correlated, that means there is positive cooperativity, then:

$$\begin{aligned} P_{1b} > P_{1u} &\Leftrightarrow \frac{a}{a+c} > \frac{b}{b+d} \Leftrightarrow \gamma = \frac{ad}{bc} > 1, \\ P_{2b} > P_{2u} &\Leftrightarrow \frac{a}{a+b} > \frac{c}{c+d} \Leftrightarrow \gamma = \frac{ad}{bc} > 1. \end{aligned} \quad [4]$$

Thus, $\gamma > 1$ is necessary and sufficient for positive cooperativity. $\gamma = +\infty$ means completely positive cooperativity ("all or none"). In nature, positive cooperativity can be caused by TF complexes constituted by protein-protein interactions (8–10).

Similarly, if Site 1 and Site 2 are negatively correlated, that means there is negative cooperativity, then:

$$\begin{aligned} P_{1b} < P_{1u} &\Leftrightarrow \frac{a}{a+c} < \frac{b}{b+d} \Leftrightarrow \gamma = \frac{ad}{bc} < 1, \\ P_{2b} < P_{2u} &\Leftrightarrow \frac{a}{a+b} < \frac{c}{c+d} \Leftrightarrow \gamma = \frac{ad}{bc} < 1. \end{aligned} \quad [5]$$

Thus, $\gamma < 1$ is necessary and sufficient for negative cooperativity. $\gamma = 0$ means completely negative cooperativity ("either-or"). In nature, negative cooperativity can arise from competitive binding among TFs with closely adjacent binding sites (11). Moreover, the cooperativity between two TFs binding at adjacent sites can oscillate between positive and negative situations with a period of ~ 10 base pairs due to DNA allostery (12, 13).

In summary, the γ coefficient ranges from 0 to positive infinity, with values indicating the degree of cooperativity (22). It is commonly known as odds ratio in statistics, which quantifies the strength of the relationship between two conditions (23). For convenience, $\log_{10} \gamma$ can also be used, with zero for no cooperativity, positive value for positive cooperativity, and negative value for negative cooperativity.

Other Statistics for the Quantification of TF Binding Cooperativity. In statistics, there is another way to quantify the correlation between the two sites due to cooperativity as follows:

$$\begin{aligned} (P_{1b} - P_{1u})(P_{2b} - P_{2u}) \\ = \left(\frac{a}{a+c} - \frac{b}{b+d} \right) \left(\frac{a}{a+b} - \frac{c}{c+d} \right). \end{aligned} \quad [6]$$

This has been defined as Φ^2 (24), and through simple derivation:

$$\Phi = \frac{ad - bc}{\sqrt{(a+b)(c+d)(a+c)(b+d)}}. \quad [7]$$

The Φ coefficient is statistically analogous to the Pearson correlation coefficient when applied to two binary events represented by Bernoulli random variables (details in [SI Appendix, Extended Methods](#)). Compared to the γ coefficient, the advantage of the Φ coefficient is that it ranges from -1 to 1 , instead of from 0 to positive infinity. It can be interpreted as 0 for no cooperativity, 1 for completely positive cooperativity (all or none), -1 for completely negative cooperativity (either-or), and nominally between 0 and 1 for partially positive cooperativity and between -1 and 0 for partially negative cooperativity. The disadvantage of the Φ coefficient is that it has TF concentration dependence, even under thermodynamic equilibrium, further explained in the next section.

With a simple derivation, dividing both the numerator and denominator in Eq. 7 by bc , the Φ coefficient can be expressed in terms of the γ coefficient as

$$\Phi = \frac{(\gamma - 1)}{\sqrt{\left(\frac{a}{b} + 1\right)\left(1 + \frac{d}{c}\right)\left(\frac{a}{c} + 1\right)\left(1 + \frac{d}{b}\right)}}. \quad [8]$$

This relationship is further illustrated in Fig. 1B, where it is discussed in a simplified scenario in which identical TFs bind on two identical sites, i.e., $b = c$. Three conditions are shown: a high binding fraction (BF , when the adjacent site is not bound) with $BF = \frac{b}{b+d} = 0.91$ ($\frac{b}{d} = 10$, blue line), a medium binding fraction with $BF = 0.5$ ($\frac{b}{d} = 1$, orange line), and a low binding fraction with $BF = 0.09$ ($\frac{b}{d} = 0.1$, pink line). Under all conditions, Φ increases monotonically with γ , and reaches saturation at very high or low values of γ .

The statistical power to significantly detect TF binding cooperativity can be theoretically illustrated through mutual information (MI , Eq. 9, $N = a + b + c + d$) from information theory (25, 26). Here, MI quantifies the binding status gained at one binding site by observing the binding status at the other site.

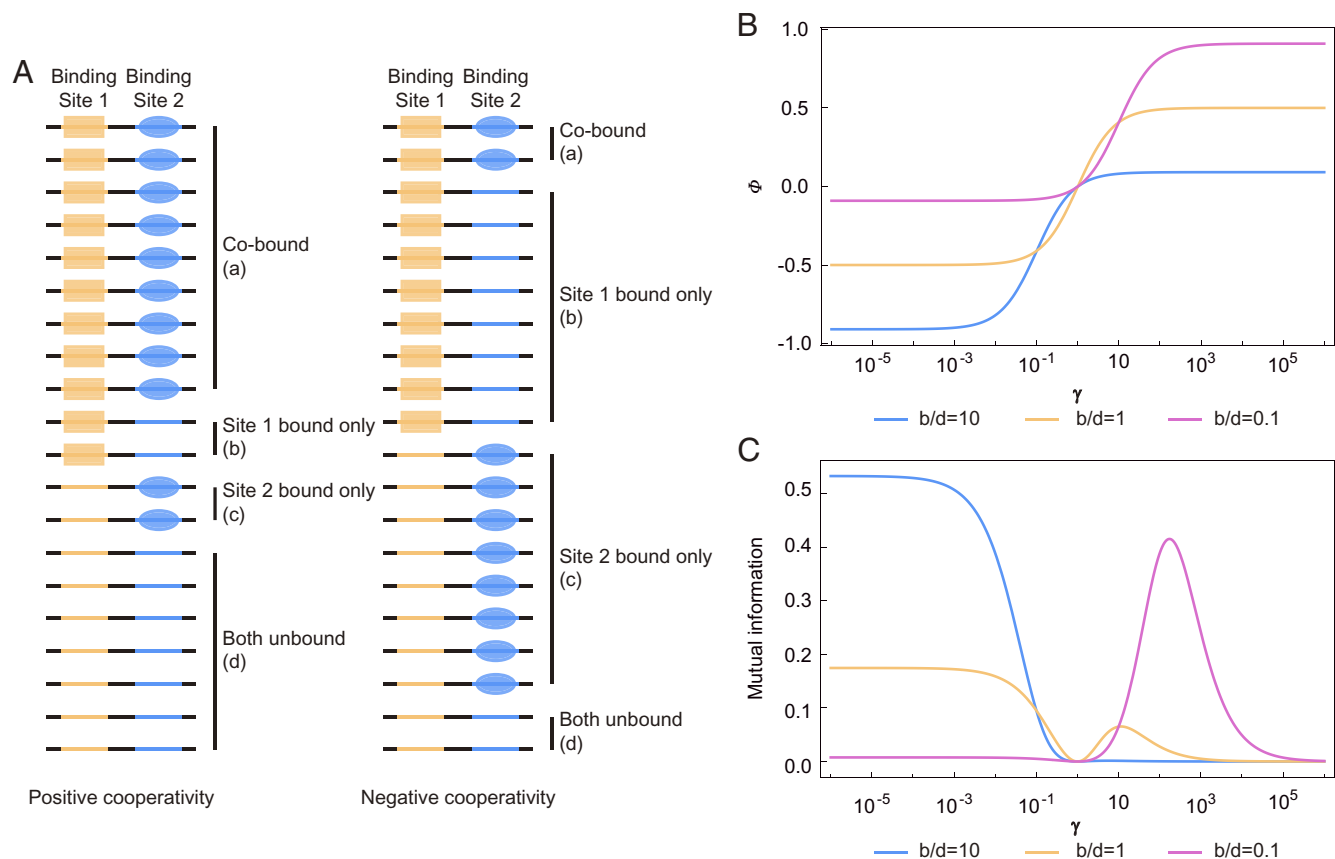


Fig. 1. Quantification of TF binding cooperativity. (A) Schematics of single molecules with two adjacent TF binding sites, showing scenarios of positive cooperativity (Left) and negative cooperativity (Right). Orange rectangle: TF A; blue oval: TF B; Black straight line: DNA. Single-molecule counts are shown as a , b , c , and d in parentheses. (B) Relationship between ϕ and γ for the case in which identical TFs bind on two identical sites ($b = c$). Three conditions are shown, each corresponding to high (blue), medium (orange), or low (pink) TF binding fraction respectively, when the adjacent site is not bound. (C) Relationship between mutual information and γ for the same case and conditions in B.

By definition, MI is nonnegative. When MI is zero, it indicates no information, due to either no cooperativity or the estimation error of cooperativity being as large as infinite, i.e., impossible to make any conclusion on TF binding cooperativity no matter how many DNA molecules are measured. A small MI implies limited information, requiring a large number of DNA molecules to detect significant cooperativity. Conversely, a high MI indicates substantial information, allowing for the detection of significant cooperativity with a relatively smaller number of DNA molecules.

$$MI = \frac{a}{N} \ln \frac{aN}{(a+b)(a+c)} + \frac{b}{N} \ln \frac{bN}{(a+b)(b+d)} + \frac{c}{N} \ln \frac{cN}{(c+d)(a+c)} + \frac{d}{N} \ln \frac{dN}{(c+d)(b+d)}. \quad [9]$$

The relationship between MI and γ is illustrated in Fig. 1C, with the same simplification and conditions in Fig. 1B. The patterns of γ in these conditions are quite complicated. The expectation is that the power to significantly detect larger $|\log_{10} \gamma|$ is larger, i.e., MI becomes larger when $|\log_{10} \gamma|$ becomes larger. However, many situations violate this expectation. For example, under the high binding fraction condition, MI is near zero for $\gamma > 1$. In the medium and low binding fraction conditions, MI becomes very small when γ becomes very large. These are due to the fact that the power to significantly detect positive cooperativity is very small when almost all the DNA molecules are cobound (b , c , and d are zero or almost zero). Conversely, under the low binding fraction condition, MI is near zero for $\gamma < 1$. This is due to the fact that the power to significantly

detect negative cooperativity is very small when almost all the DNA molecules are unbound at both sites (a , b , and c are zero or almost zero).

TF Binding Cooperativity Under Thermodynamic Equilibrium.

In the preceding discussion, we have demonstrated TF binding cooperativity under probability and statistics perspectives. Here, we focus on elucidating TF binding cooperativity under thermodynamic equilibrium. We present a thermodynamic model describing the binding of two TFs at adjacent sites on DNA (Fig. 2A). For each reversible reaction, we define k_+ and k_- as the kinetic rate constants for the forward binding and reverse dissociation reactions, respectively. When this system reaches thermodynamic equilibrium, the forward binding reaction is balanced by the reverse dissociation reaction, and the binding affinity represented by the association binding affinity $K_{i \rightarrow j} = \frac{k_+}{k_-}$ (i, j stand for different states) is defined as follows ($[A]$ or $[B]$ represents the concentration of TF A or TF B):

$$\begin{aligned} K_{0 \rightarrow 1} &= \frac{b}{d \cdot [A]}, \\ K_{0 \rightarrow 2} &= \frac{c}{d \cdot [B]}, \\ K_{1 \rightarrow 3} &= \frac{a}{b \cdot [B]}, \\ K_{2 \rightarrow 3} &= \frac{a}{c \cdot [A]}. \end{aligned} \quad [10]$$

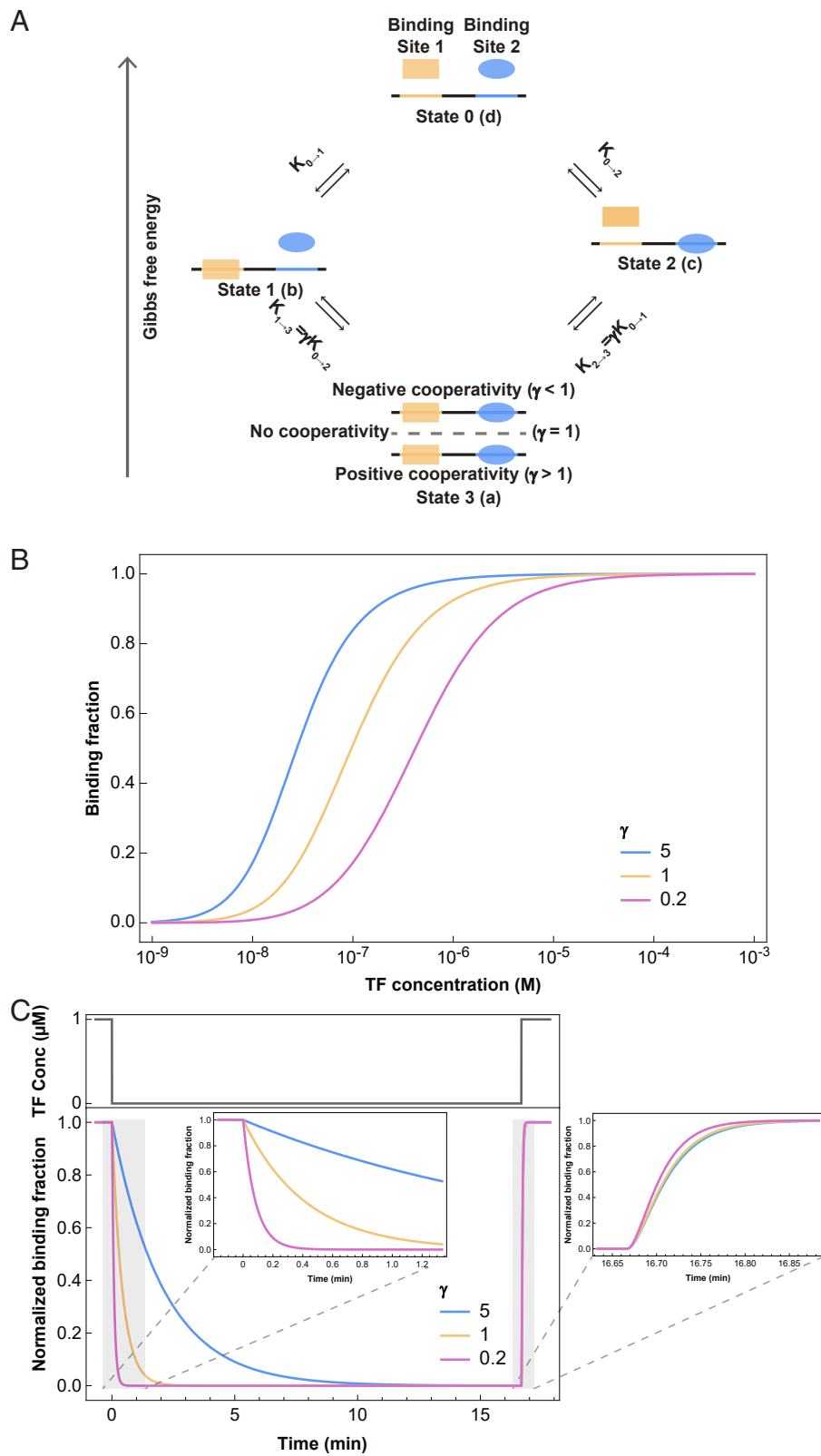


Fig. 2. Thermodynamics and kinetics of two TFs' binding at adjacent DNA sites. (A) Thermodynamic cycle illustrating the binding of two TFs at adjacent DNA sites (adapted from figure 4C in ref. 7). Four possible DNA binding states (0 to 3) are depicted, with single-molecule counts shown in parentheses. Binding affinities are marked, where positive cooperativity enhances the binding affinity ($\gamma > 1$) and negative cooperativity reduces it ($\gamma < 1$). Orange rectangle: TF A; blue oval: TF B; black straight line: DNA. (B) Cobinding fraction (a special case of State 3 in A, supposing identical TFs binding on two adjacent sites with identical binding affinities) exhibits sigmoidal dependence of TF concentration for different γ values under thermodynamic equilibrium. (C) Hysteretic behavior of cobinding fraction (the same case as B) to oscillating TF concentration in the form of a rectangular wave. *Top panel*: TF concentration oscillates between 1 μM and 0 as a rectangular wave with a duty cycle of 50% and a period of 1,000 s. *Left Inset*: enlarged view showing the cobinding fraction decreases at varying rates with different γ values as the TF concentration suddenly drops down from 1 μM to 0. *Right Inset*: enlarged view showing the cobinding fraction ascends at varying rates with different γ values as the TF concentration shifts from 0 to 1 μM .

Given that Gibbs free energy is a state function and thus independent of the reaction pathway, we have

$$\Delta G_{0 \rightarrow 1}^\circ + \Delta G_{1 \rightarrow 3}^\circ = \Delta G_{0 \rightarrow 2}^\circ + \Delta G_{2 \rightarrow 3}^\circ, \quad [11]$$

where $\Delta G_{i \rightarrow j}^\circ = -RT \ln K_{i \rightarrow j}$; thus we obtain

$$K_{0 \rightarrow 1} \cdot K_{1 \rightarrow 3} = K_{0 \rightarrow 2} \cdot K_{2 \rightarrow 3}. \quad [12]$$

This thermodynamic cycle has also been previously discussed (12). Here, we find the γ coefficient defined above is actually the change of the binding affinities as

$$\frac{K_{2 \rightarrow 3}}{K_{0 \rightarrow 1}} = \frac{K_{1 \rightarrow 3}}{K_{0 \rightarrow 2}} = \frac{ad}{bc} = \gamma. \quad [13]$$

Clearly, the γ coefficient is independent of TF concentrations.

When $\gamma > 1$, this declares the binding affinities of TF A and TF B increase together (positive cooperativity). The existence of positive cooperativity causes relatively lower Gibbs free energy (more stable, Fig. 2A). Conversely, when $\gamma < 1$, the binding affinities of TF A and TF B decrease together (negative cooperativity). Negative cooperativity results in relatively higher Gibbs free energy (less stable, Fig. 2A). This has also been addressed by Wyman, based on the principles of equilibrium thermodynamics (27). Compared to the widely used Hill coefficient, γ provides a more intuitive and perceivable assessment for binding cooperativity between two TFs.

With a simple derivation, substituting a, b, c, d with TF concentrations and binding affinities in Eq. 8 (details in [SI Appendix, Extended Methods](#)), Φ coefficient can be expressed in terms of γ , TF concentrations, and binding affinities as follows:

$$\Phi = \frac{\gamma - 1}{\sqrt{(1 + \gamma[A]K_{0 \rightarrow 1}) \left(1 + \frac{1}{[A]K_{0 \rightarrow 1}}\right)}} \times \frac{1}{\sqrt{(1 + \gamma[B]K_{0 \rightarrow 2}) \left(1 + \frac{1}{[B]K_{0 \rightarrow 2}}\right)}}. \quad [14]$$

It is evident that the Φ coefficient is influenced not only by variations in binding affinity (γ) but also by TF concentrations and binding affinities. Therefore, γ serves as an effective measure of variations in TF binding affinity due to cooperativity, while Φ is well suited for capturing TF binding fraction variations due to cooperativity.

Functional Relevance of Thermodynamic Positive Cooperativity.

After illustrating TF binding cooperativity under thermodynamic equilibrium, now we are focusing on revealing their functional relevance. We first study the functional relevance of positive cooperative binding under thermodynamic equilibrium. To simplify the system, we assume that the two TFs and their binding sites in Fig. 2A are identical. The simulation is carried out under these parameters: $k_{+,0 \rightarrow 1} = 5 \times 10^5 \text{ M}^{-1}\text{s}^{-1}$, $k_{-,1 \rightarrow 0} = 0.02 \text{ s}^{-1}$, and $K_D = \frac{k_{-,1 \rightarrow 0}}{k_{+,1 \rightarrow 0}} = 40 \text{ nM}$, which are close to the TF binding in mammalian cells and consistent with the literature (28–30). Fig. 2B illustrates the response of the cobinding fraction (State 3 in Fig. 2A) to TF concentration with different γ values. All response curves follow a sigmoidal pattern, in which cobinding with positive cooperativity ($\gamma > 1$) exhibits a more rapid response to changes in concentration, thereby acting as a sensitive response around a threshold of TF concentration.

Functional Relevance of Thermodynamic Negative Cooperativity. Negative cooperativity implies that the binding of TF A would weaken the binding of TF B, resulting in an either-or tendency for TF A and B to bind DNA. Consequently, negative cooperativity ensures mutual exclusivity in TF binding, akin to the CI-CRO system in phage λ (6), and facilitates bidirectional gene switching.

In living organisms, TF concentrations often oscillate in response to specific cellular and environmental stimuli (31, 32). Under the assumption that TF–DNA binding reaches thermodynamic equilibrium, fluctuations of TF concentrations fundamentally represent a transition from one thermodynamic equilibrium state to another within this TF–DNA binding system, wherein we examine the functional relevance of negative cooperative binding. To address this challenge, we commence with a simplified model of Fig. 2A which takes A and B as the same TF and assumes two identical binding sites. These assumptions signify that the two binding sites share the same binding affinity ($K_A = \frac{k_+}{k_-}$), association rate constant (k_+), and dissociation rate constant (k_-). In this scenario, we could derive three time-dependent differential equations with specific boundary conditions (details in [SI Appendix, Extended Methods](#)).

To obtain a concise physicochemical model, we introduce several additional assumptions. First, we simply take the change of TF concentration as a rectangular wave function. Second, although both the association rate constant (k_+) and dissociation rate constant (k_-) may vary with cooperativity, we neglect changes in k_+ , as it is primarily governed by bimolecular diffusion (33). Thus, we attribute changes in association binding affinity entirely to alterations in k_- .

Hysteretic behavior (34) of TF cobinding fraction (Fig. 2A, State 3) to TF concentration oscillation is simulated under these kinetic parameters: $k_{+,0 \rightarrow 1} = 5 \times 10^5 \text{ M}^{-1}\text{s}^{-1}$, $k_{-,1 \rightarrow 0} = 0.02 \text{ s}^{-1}$, [A] is treated as a rectangular wave oscillating between 1 μM and 0 with a duty cycle of 50% and a period of 1,000 s, and total DNA concentration is normalized to 1.

Fig. 2C illustrates the hysteretic behavior of TF cobinding fraction to oscillating TF concentration in the form of a rectangular wave. The simulation starts from thermodynamic equilibrium binding states (binding fractions of different γ values are normalized to 1). For $\gamma = 0.2$ (negative cooperativity), the binding fraction declines rapidly, reaching the bottom within 40 s (*Left Inset* of Fig. 2C), suggesting a fast response to TF concentration fluctuation. In contrast, when $\gamma = 5$, the cobinding fraction for positive cooperativity reaches the bottom much slower than that for negative cooperativity. The *Right Inset* of Fig. 2C provides a zoom-in view of the hysteretic behavior of cobinding fraction when the TF concentration rises from 0 to 1 μM as a step function, due to rapid biomolecular binding.

In summary, negative binding cooperativity enables either mutual exclusion of TFs similar to the CI-Cro system in phage λ (6) or fast TF dissociation from the DNA in response to rapid intranuclear TF concentration changes, hence a rapid response of transcription.

Cooperative Binding of Three TFs. In eukaryotic cells, multiple TFs bind to adjacent sites within DNA regulatory regions to regulate gene expression (7, 10). The analyses above focus on two TFs' binding on DNA. In this section, we extend the discussion to the case of three TFs' binding at adjacent DNA sites and investigate whether the conclusions drawn from two TFs' binding are applicable to three TFs' binding.

Due to the third TF's binding introduced into the system, the number of possible binding states increases to $2^3 = 8$, making

the system much more complicated. The thermodynamic cycles of three TFs' binding are illustrated in Fig. 3A. To simplify the analysis, we bring in a new assumption that the binding of TF C and the binding of TF B are independent of each other, as they are not adjacent to each other. This simplifies the system to two distinct cooperativity parameters, γ_{CA} and γ_{AB} . Under thermodynamic laws, the relationship between the binding affinities can be expressed as follows (details in [SI Appendix, Extended Methods](#)):

$$\begin{aligned}\gamma_{CA} &= \frac{K_{1 \rightarrow 4}}{K_{0 \rightarrow 2}}, \\ \gamma_{AB} &= \frac{K_{2 \rightarrow 6}}{K_{0 \rightarrow 3}}, \\ \frac{K_{1 \rightarrow 5}}{K_{0 \rightarrow 3}} &= \frac{K_{4 \rightarrow 7}}{K_{2 \rightarrow 6}} = 1.\end{aligned}\quad [15]$$

Therefore, we have

$$\gamma_{total} = \frac{K_{5 \rightarrow 7}}{K_{0 \rightarrow 2}} = \gamma_{CA} \cdot \gamma_{AB}. \quad [16]$$

If we set $\gamma_{CA} = \gamma_{AB} = \gamma$, then we have $\gamma_{total} = \gamma^2$. We performed the numerical simulation with this restriction and use the same assumptions and kinetics parameters as those for two TFs' binding. Fig. 3B shows the steeper sigmoidal concentration dependence of the three TFs' cobinding fraction (Fig. 3A, State 7) than the two TFs'. Hysteretic behavior of cobinding fraction in response to a rectangular wave function of TF concentration, with different γ values, is shown in Fig. 3C with an analogous conclusion to that of Fig. 2C, with an even faster response time from the equilibrium value than two TFs' case due to the third TF's binding.

Binding Cooperativity Under Nonequilibrium Steady States.

In the previous sections, we have discussed TF binding cooperativity and their functional relevance under thermodynamic equilibrium. However, in living cells, TF binding and other regulatory protein binding can also be affected by free energy driven enzymatic processes, such as ATP hydrolysis, and reach nonequilibrium steady states (NESSs) (16, 35). Here, we introduce two models under NESSs with the hypothesis that there is no cooperative binding under the thermodynamic equilibrium, and examine how binding cooperativity arise in such systems due to NESSs.

ATP-dependent chromatin remodeling processes play a crucial role in modulating chromatin structure, thus influencing the accessibility of DNA to various proteins, including TFs. The remodeling complexes guided by pioneer factors, such as SWI/SNF, utilize energy from ATP hydrolysis to reposition nucleosomes, facilitating the binding of TFs to DNA (15). The negative cooperative binding between a nucleosome and a remodeling complex (we simplify the combination of a remodeling complex and a pioneer factor as a remodeling complex here and hereafter) can be simplified as the NESS model shown in Fig. 4A (36). The $k_{+,i \rightarrow j}$ and $k_{-,j \rightarrow i}$ are the association and dissociation rate constants from State i to State j , respectively. The $k'_{+,i \rightarrow j}$ and $k'_{-,j \rightarrow i}$ are the ADP/ATP-coupled association and dissociation rate constants from i to j state.

Under NESS, we have

$$\frac{dP(S_i)}{dt} = 0. \quad [17]$$

$P(S_i)$ is the time-dependent probability of each DNA state. The net flux J is defined as follows (where CCW and CW denote counterclockwise and clockwise, respectively, and $[A]$ represents the concentration of nucleosome):

$$\begin{aligned}J &= J_+ - J_- = J_{CCW} - J_{CW} \\ &= k_{+,0 \rightarrow 1}P(S_0)[A] - k_{-,1 \rightarrow 0}P(S_1).\end{aligned}\quad [18]$$

When the net flux $J > 0$, i.e., a counterclockwise net flux in the NESS system in Fig. 4A, the remodeling complex continuously exerts force to reposition the nucleosome. Under this assumption together with some other hypotheses, we can prove that $\gamma < 1$ (details in [SI Appendix, Extended Methods](#)), which implies that the nucleosome and the remodeling complex exhibit a mutually exclusive pattern, i.e., negative cooperativity).

Furthermore, we examine the monotonicity of γ as a function of ATP or ADP concentrations. Intuitively, this remodeling process depends on the concentrations of ATP or ADP. When the ATP concentration increases, the reaction flux shifts toward promoting nucleosome repositioning. In contrast, when the ADP concentration rises, the reaction flux is weakened. We prove that the γ coefficient decreases monotonically as ATP concentration increases or ADP concentration decreases, indicating a shift toward greater negative cooperativity (details in [SI Appendix, Extended Methods](#)).

To intuitively illustrate the relationship between the γ coefficient and the ATP/ADP concentration ratio, we also provide numerical solutions by introducing some hypotheses and a series of parameters as follows. First, we assume that there is no cooperativity in the corresponding thermodynamic system, which leads to:

$$\begin{aligned}k_{+,0 \rightarrow 2} &= k_{+,1 \rightarrow 3} = k_+, \\ k_{-,2 \rightarrow 0} &= k_{-,3 \rightarrow 1} = k_-.\end{aligned}\quad [19]$$

The rate constants of each reaction are estimated from in vitro experimental results (37–40). Second, to simplify the calculations, we incorporate the effects of the nucleosome and remodeling complex concentrations into the rate constants, thereby allowing us to focus solely on a few rate constants with units of s^{-1} or $M^{-1}s^{-1}$. The simulation is carried out under these kinetic parameters: $k_- = 0.02 s^{-1}$, $k'_+ = k_+[B] = 0.03 s^{-1}$, $k_{-,1 \rightarrow 0} = k_{-,3 \rightarrow 2} = 0.001 s^{-1}$, $k_{+,0 \rightarrow 1}[A] = k_{+,2 \rightarrow 3}[A] = 0.5 s^{-1}$, $k'_{-,3 \rightarrow 2} = 1,000 M^{-1}s^{-1}$, $k'_{+,2 \rightarrow 3}[Pi][A] = 0.015 M^{-1}s^{-1}$. Intracellular phosphate (Pi) concentration remains stable at the millimolar level, so we set $[Pi] = 3 \text{ mM}$ (41). Fig. 4B demonstrates how the γ coefficient varies with the ATP/ADP concentration ratio, clearly showing that $\gamma < 1$ as described by the model in Fig. 4A, and the γ coefficient decreases monotonically as ATP/ADP concentration ratio increases, which is consistent with the theoretical analysis.

Similarly, the positive cooperative binding between a TF (binding to the newly accessible region due to remodeling) and a remodeling complex can be simplified as the NESS model shown in Fig. 4C (42). We define the net flux J as follows (note that the definition of $[A]$ and directions of J_+ and J_- in this model differ from those in the model presented in Fig. 4A. $[A]$ represents the concentration of a TF in this model. Other definitions remain unchanged.):

$$\begin{aligned}J &= J_+ - J_- = J_{CW} - J_{CCW} \\ &= k_{-,1 \rightarrow 0}P(S_1) - k_{+,0 \rightarrow 1}P(S_0)[A].\end{aligned}\quad [20]$$

When the net flux $J > 0$, i.e., a clockwise net flux in the NESS system in Fig. 4C, the remodeling complex continuously exerts

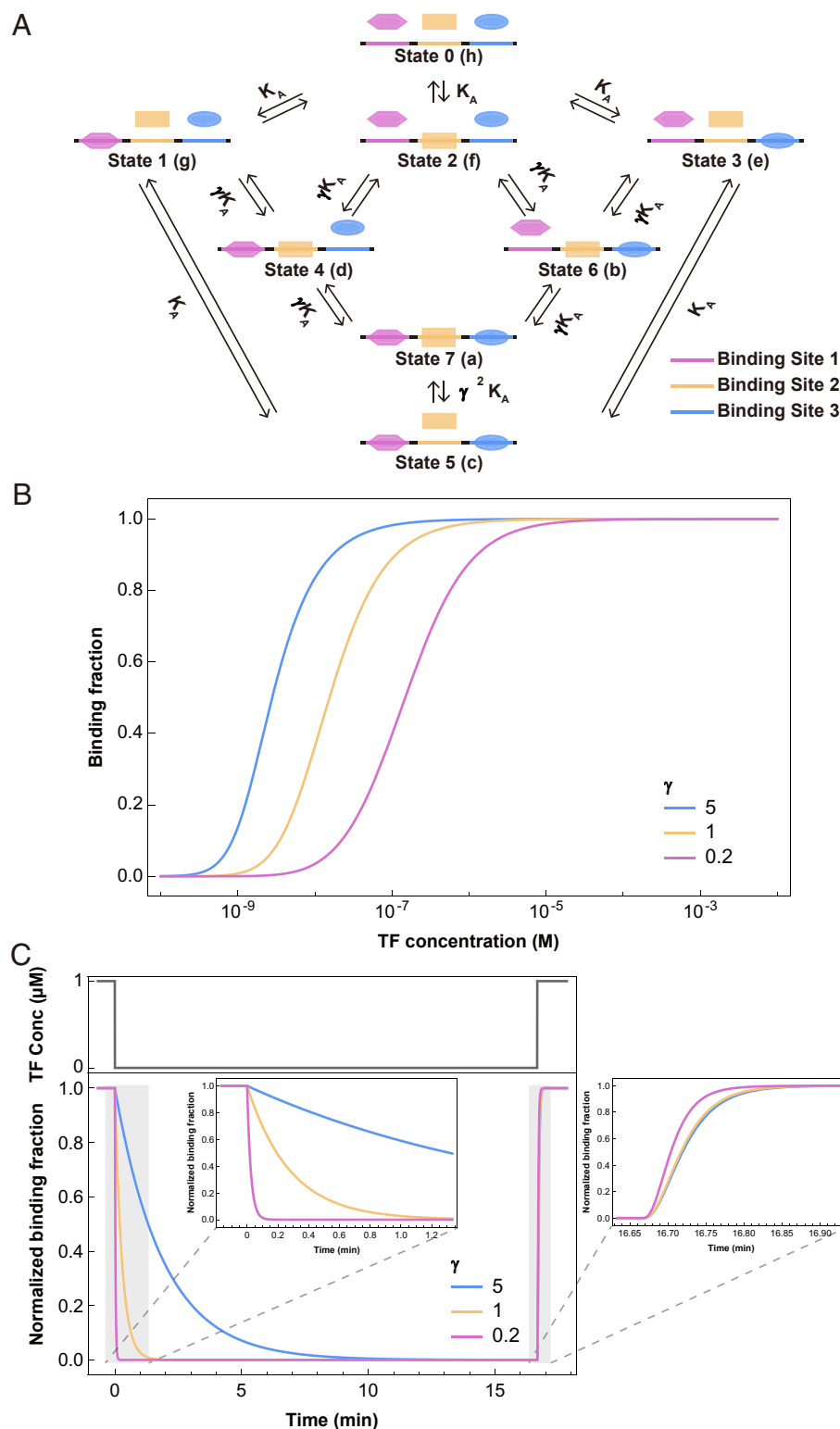


Fig. 3. Thermodynamics and kinetics of three TFs' binding at adjacent sites on DNA. For simplicity, we set three identical TFs binding at three adjacent DNA sites with identical binding affinities. In addition, we also set $\gamma_{CB} = 1$ and $\gamma_{CA} = \gamma_{AB} = \gamma$. (A) Schematics of thermodynamic cycles of three TFs' (C, A, and B) binding at adjacent sites on DNA. Eight DNA binding states (0 to 7) are presented (corresponding single molecule counts are shown in parentheses, which are used for the thermodynamics and kinetics derivation in *SI Appendix, Extended Methods*). Binding affinities are marked. Orange rectangle: TF A; blue oval: TF B; pink hexagon: TF C; black straight line: DNA. (B) Dependence of the cobinding fraction (State 7 in A) on TF concentration, exhibiting sigmoidal response for various γ values. (C) Hysteretic behavior of cobinding fraction (State 7 in A) to oscillating TF concentration in the form of a rectangular wave. *Top panel*: TF concentration oscillates between 1 μ M and 0 as a rectangular wave with a duty cycle of 50% and a period of 1,000 s. *Left Inset*: enlarged view showing the cobinding fraction decreases at varying rates with different γ values as the TF concentration suddenly drops down from 1 μ M to 0. *Right Inset*: enlarged view showing the cobinding fraction ascends at varying rates with different γ values as the TF concentration shifts from 0 to 1 μ M.

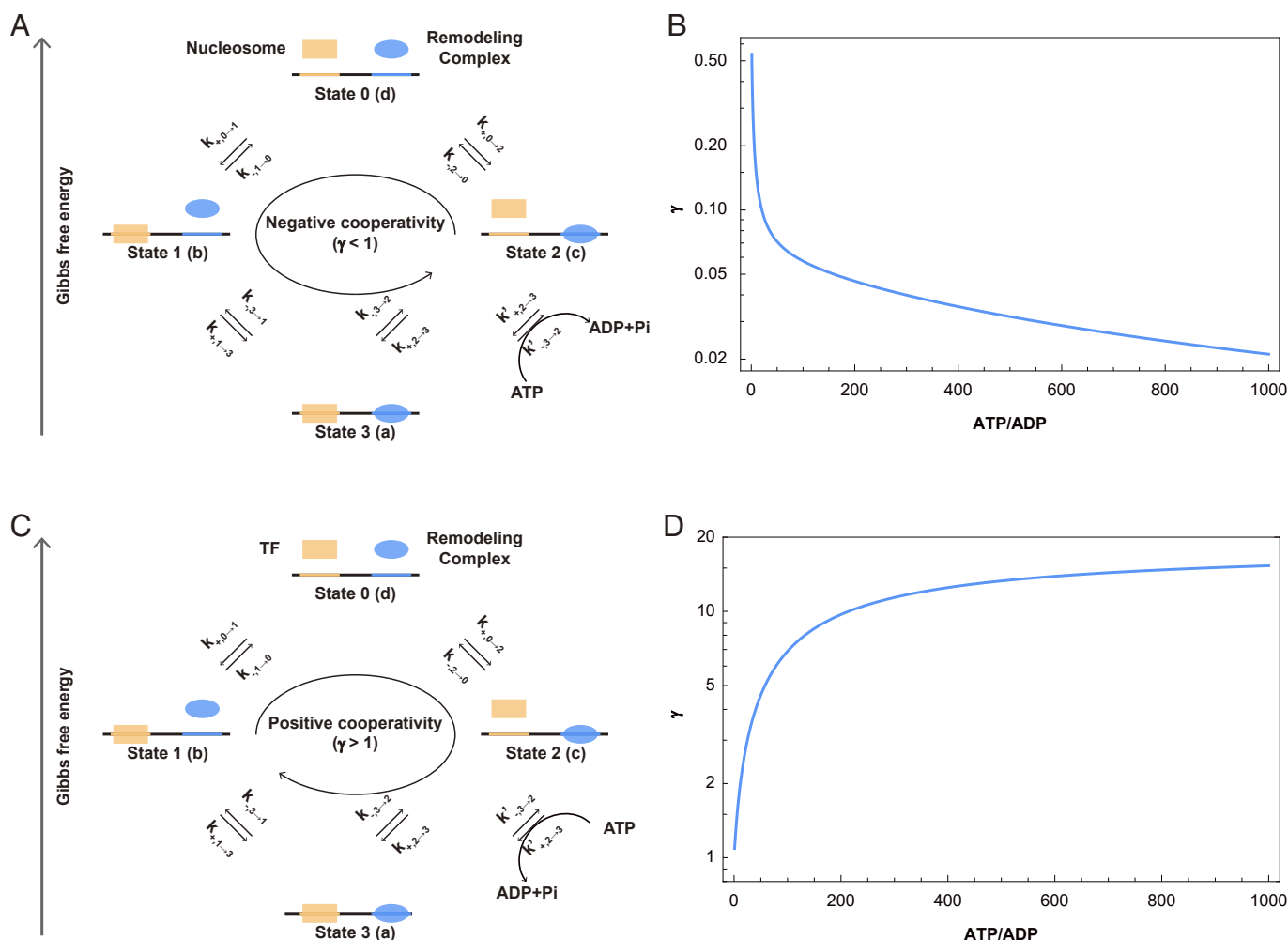


Fig. 4. Binding models and related simulation under nonequilibrium steady states with negative/positive cooperativity, respectively. Four possible DNA binding states (0 to 3) are depicted, with single-molecule counts shown in parentheses. The $k_{+/-i \rightarrow j}$ or $k'_{+/-i \rightarrow j}$ are the association or dissociation rate constants from State i to State j without or with ATP/ADP coupling, respectively. Orange rectangle: nucleosome or TF; blue oval: remodeling complex; black straight line: DNA. (A) Nonequilibrium steady state depicting ATP-dependent chromatin remodeling process (negative cooperativity between nucleosome and remodeling complex, $\gamma < 1$). (B) Relationship between the γ coefficient and ATP/ADP concentration ratio corresponding to the model in A. The simulation result shows a monotonic decrease of the γ coefficient with increasing ATP/ADP ratio, aligning with theoretical analysis. (C) Nonequilibrium steady state with positive cooperativity between TF and remodeling complex ($\gamma > 1$). (D) Relationship between the γ coefficient and ATP/ADP concentration ratio corresponding to the model in C. The simulation result shows a monotonic increase of the γ coefficient with increasing ATP/ADP ratio, which is consistent with theoretical analysis.

force to facilitate other TF's binding. Under this assumption together with some other hypotheses, we can prove that $\gamma > 1$ (details in *SI Appendix, Extended Methods*), which implies that the TF and the remodeling complex exhibit a cobinding pattern, i.e., positive cooperativity. We also further prove that the γ coefficient increases monotonically as ATP concentration increases or ADP concentration decreases, indicating a shift toward greater positive cooperativity (details in *SI Appendix, Extended Methods*).

Fig. 4D shows the relationship between the γ coefficient and the ATP/ADP concentration ratio based on the model presented in Fig. 4C. Most of the kinetic parameters used in the simulation are the same as those in Fig. 4B, with some modifications due to differences in the models, which are listed as follows: $k_{-,3 \rightarrow 2} = 0.5 \text{ s}^{-1}$, $k_{+,0 \rightarrow 1}[A] = k_{+,2 \rightarrow 3}[A] = 0.0001 \text{ s}^{-1}$, $k'_{+,2 \rightarrow 3} = 1 \times 10^7 \text{ M}^{-2} \text{ s}^{-1}$. Fig. 4D also clearly shows that $\gamma > 1$ as described by the model in Fig. 4C, and the γ coefficient increases monotonically as ATP/ADP concentration ratio increases, which is consistent with the theoretical analysis.

In summary, NESS can lead to either positive or negative cooperativity between two binding sites even when there is no cooperativity between these two sites under thermodynamic equilibrium, and its quantification can also be indicated by the γ coefficient.

Conclusions

We introduce three metrics to quantify the binding cooperativity between two adjacent TFs: the γ coefficient, the Φ coefficient, and mutual information (MI). Among these metrics, the Φ coefficient captures the correlation between overall binding fractions, but is dependent on TF concentrations. MI quantifies how the binding of one site can provide information for the other, though it does not reflect the strength of cooperativity. Thermodynamically, the γ coefficient is independent of TF concentrations and can serve as an intrinsic metric for TF binding cooperativity under thermodynamic equilibrium. Therefore, we suggest that γ is the optimal metric for quantifying TF binding cooperativity. Thermodynamic analysis of γ demonstrates that,

the binding affinities of the two TFs at adjacent sites either increase together due to positive cooperativity or decrease together due to negative cooperativity, rather than one increasing while the other decreasing (27). Our thermodynamic investigation further reveals that the functional significance of positive cooperativity lies in its ability to produce sigmoidal dependence of the cobinding fraction on TF concentration, thereby enhancing sensitivity. This behavior mirrors the classic receptor–ligand positive cooperativity observed in the hemoglobin–oxygen interaction (1). Given that TF concentrations ubiquitously oscillate in the cell nucleus (31, 32), our kinetics analysis highlights the functional significance of negative cooperativity in assuring rapid TF dissociation from DNA in response to abrupt changes in TF concentration, hence rapid switching of transcription. Furthermore, we demonstrated that the nonequilibrium steady states of living cells can lead to either positive or negative cooperativity, and its quantification can also be indicated by the γ coefficient.

Materials and Methods

The relationship between Φ and γ , as well as the equivalence of Φ to the Pearson correlation coefficient, is derived in *SI Appendix, Extended Methods*.

The thermodynamic cycles for the binding of two and three TFs at adjacent DNA sites are also derived, with the former presented in the Results section and the latter in *SI Appendix, Extended Methods*. The differential equations of the kinetics of two and three TFs' binding to adjacent sites on DNA are shown in *SI Appendix, Extended Methods*. The derivation of γ under nonequilibrium steady states is provided in *SI Appendix, Extended Methods*. All analytical and numerical solutions, as well as the corresponding plots, were generated using Mathematica (version 13.0).

Data, Materials, and Software Availability. All study data are included in the article and/or *SI Appendix*.

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